Bilingual Article-Aware Retrieval-Augmented Generation for ANVISA Regulatory Document Interpretation

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Abstract

Background. Regulatory agencies publish long and terminology-dense documents that are difficult to search across languages. Retrieval-augmented generation may improve access to regulatory information, but use in this setting requires source traceability and refusal when evidence is insufficient.

Objective. To develop and technically evaluate a bilingual Portuguese-English retrieval-augmented generation system for selected ANVISA pharmaceutical and health-product regulatory documents.

Methods. Raw regulatory text files were cleaned and indexed as article-aware chunks with document, article, and citation metadata. Retrieval combined dense vector search and BM25, merged results by reciprocal rank fusion, reranked candidate chunks, and generated answers from the top five contexts. English questions were expanded with a fixed Portuguese-English regulatory glossary. A 200-question bilingual benchmark tested answerable, multi-article, exception, unanswerable, and adversarial questions. Automated evaluation measured retrieval, citation validity, answer completeness, faithfulness, and refusal behavior.

Results. The final index contained 382 articles and 481 chunks from six ANVISA documents. On the post-repair internal held-out benchmark, article Recall@5 was 1.000 in both languages. AI-judge completeness was 0.995 in English and 1.000 in Portuguese, with refusal correctness of 1.000. A prospective 50-question external stress test showed normalized article Recall@5 of 0.967 and answer completeness of 0.943, but refusal correctness was lower at 0.700. After refusal calibration on separate development data, a new 30-question refusal-focused test achieved refusal correctness of 0.944, with no over-refusal on answerable controls.

Conclusion. Article-aware indexing, hybrid retrieval, reranking, bilingual query expansion, and strict refusal rules supported technical validation of a bilingual ANVISA regulatory-document

RAG system. The findings do not establish regulatory advice, expert consensus, or deployment readiness.

Introduction

Regulatory agencies publish information that guides pharmaceutical and health-product development, submission, monitoring, and post-authorization obligations. These materials are often long, article-structured, and dependent on exact terminology. A user may need to locate a specific article, distinguish a general rule from an exception, or determine whether a requirement applies to a clinical trial, biological product, medical device, or stability study. In Brazil, many health-product regulatory documents are published in Portuguese. This creates a practical barrier for international scientists, sponsors, and companies that need to work with Portuguese source material but may ask questions in English.

Retrieval-augmented generation (RAG) combines document retrieval with language-model generation and has been proposed as a way to reduce unsupported responses in knowledge-intensive question answering [1]. In regulatory use, however, the question is not only whether an answer is fluent. A useful system must retrieve the correct source document, preserve article-level provenance, cite the evidence used, and refuse questions that cannot be answered from the indexed corpus. These requirements are stricter than ordinary chatbot behavior because unsupported answers can mislead users who are trying to interpret regulatory obligations.

Recent work in Therapeutic Innovation & Regulatory Science evaluated a RAG-based generative AI tool for semantic search of FDA guidance documents [2]. That study showed potential for accelerating access to guidance information, but also reported incomplete answers and answers containing incorrect information. The authors concluded that correct source citation may help users locate relevant documents, while further refinement of chunking, embeddings, prompt engineering, and validation would be needed before such systems could be relied on to answer regulatory questions. This prior work provides a relevant regulatory-science precedent and highlights the need for task-specific validation.

The present study evaluates a bilingual RAG system for selected ANVISA regulatory documents. The contribution is technical and bounded. The system was designed for a closed document collection, with article-aware chunking, hybrid lexical and dense retrieval, cross-encoder reranking, Portuguese-English query expansion, citation-constrained answer generation, and refusal behavior when evidence was insufficient. The objective was to test

whether this design could support article-level retrieval and answer generation for Portuguese and English questions while maintaining transparent limits.

Materials and Methods

Study design

This was a technical validation study of a closed-corpus RAG system. The project followed a staged development and validation sequence. Initial testing identified environment, corpus-cleaning, retrieval, answer-completeness, citation, and refusal failures. Repairs were made using development data and followed by re-evaluation. Because internal held-out metrics became very high after targeted repairs, a prospective external stress test was run once before further tuning. That test showed strong answerable-question performance but weaker refusal behavior. Refusal calibration was then updated using a separate development set, and a new refusal-focused external test was run. The final pipeline was frozen after this step.

Corpus

The corpus consisted of six selected ANVISA regulatory text files relevant to clinical trials, biological products, medical devices, pharmacovigilance, and stability requirements. The indexed corpus was treated as closed. The system was not allowed to answer from current web pages, private submission systems, forms, fee schedules, unpublished information, or documents outside the index.

Table 1. Indexed ANVISA corpus.

Source file	Main topic	Cleaned characters	Chunks
IN_338_2024.txt	Normative instruction linked to RDC 945/2024	14,450	19
RDC_406_2020_Estudos_de_Estabilidade.txt	Stability studies	39,879	61
RDC_55_2010_Registro_de_Produtos_Biologicos.txt	Biological products	43,062	69
RDC_751_2022_Regras_de_classificacao_de_risco.txt	Medical-device risk classification	80,024	102
RDC_945_2024.txt	Clinical-trial regulation	71,857	123
rdc09_2015.txt	Clinical-trial regulation	53,346	107

Text cleaning and article-aware chunking

Raw files contained a mix of plain text and HTML-derived material. The cleaning pipeline removed markup and boilerplate where possible, normalized text, and saved cleaned files before indexing. The cleaned corpus was then segmented using article-aware rules. The target unit was one regulatory article per chunk, with nearby section or chapter headings preserved

when available. Long articles were split into smaller subchunks while retaining the same article citation metadata.

The final article-aware index used sentence-transformers/all-MiniLM-L6-v2 embeddings with maximum length 512. The chunking configuration used a maximum of 500 tokens for single-article chunks. Long articles were split into 400-token chunks with 60-token overlap. The final index contained 382 articles and 481 chunks. Of these, 359 articles were represented as a single chunk and 23 long articles were split into multiple chunks. Median chunk length was 155 tokens, maximum length was 503 tokens, and no chunk exceeded 700 tokens.

Retrieval architecture

The retrieval pipeline used four stages. First, dense vector retrieval selected semantic candidates. Second, BM25 selected lexical candidates over passage text and regulatory metadata. Third, reciprocal rank fusion merged dense and BM25 results. Fourth, a multilingual cross-encoder reranker sorted the fused candidates. The system retrieved 100 dense candidates and 100 BM25 candidates, retained 100 fused candidates, and passed the top five reranked contexts to answer generation. Reciprocal rank fusion used a smoothing constant of 60. The reranker was cross-encoder/mmarco-mMiniLMv2-L12-H384-v1.

Hybrid retrieval was used because regulatory questions often depend on exact terms, acronyms, document numbers, and article labels. Examples include DDCM, RDC 945, IN 338, Art. 33, sponsor, clinical trial, biological product, medical device, and stability study. Dense retrieval can match meaning across paraphrases, but lexical retrieval is important when a specific regulatory term or document number determines the correct answer.

English query expansion

English questions were expanded for retrieval using a fixed Portuguese-English regulatory glossary. The original English question was preserved for user intent and answer language. Portuguese retrieval anchors were added only to improve matching against Portuguese source documents. The glossary preserved terms such as clinical trial/ensaio clinico, sponsor/patrocinador, biological product/produto biologico, medical device/dispositivo medico, and stability study/estudo de estabilidade. Query expansion was evaluated during development with and without rewrite to confirm that it improved English retrieval.

Answer generation and refusal policy

Generated answers were constrained to the top five retrieved contexts. The answer language followed the user question. The prompt required every material statement to cite the retrieved source document and article using the stored citation label. The prompt also prohibited invented document numbers, articles, sections, deadlines, or requirements.

A strict refusal rule was added after failure analysis. If retrieved context was related but did not contain the specific requested answer, the system was instructed to return a refusal sentence without citation. English refusals used: "The retrieved documents do not provide enough information to answer." Portuguese refusals used: "Os documentos recuperados nao fornecem informacao suficiente para responder." The rule explicitly covered current forms, exact URLs, current fees, private contract terms, company-specific submission status, personal contact information, market prices, reimbursement, taxes, patents, future approvals, and advice not stated in the retrieved regulatory text.

Benchmark

The main benchmark contained 200 bilingual, article-level questions: 70 Portuguese answerable questions, 70 English answerable questions, 20 multi-article or exception questions, and 40 unanswerable or adversarial questions. Questions were designed to test source retrieval, article retrieval, completeness, citation behavior, and refusal behavior. The held-out evaluation used 75 English rows and 75 Portuguese rows. Benchmark manifests and checksums were saved to support reproducibility.

External testing used two additional benchmarks. The first was a 50-question prospective stress test with 25 bilingual pairs, including 40 answerable and 10 unanswerable or adversarial questions. The second was a 30-question refusal-focused external test with 18 unanswerable or current-specific questions and 12 answerable controls.

Evaluation metrics

Retrieval was evaluated with source hit rate, source mean reciprocal rank (MRR), article Recall@5, article Precision@5, article MRR@5, article nDCG@5, and retrieved gold-term coverage. Generation was evaluated with answer gold-term coverage, citation presence, article citation presence, retrieved-citation validity, and refusal accuracy. Automated answer-quality judging scored faithfulness, citation correctness, completeness, and refusal correctness.

Pair-clustered bootstrap confidence intervals were used where applicable so bilingual counterpart questions were not treated as independent observations. For the external

prospective test, normalized article metrics were used because article labels in the external gold file used a different format from the internal runner fields.

Automated judging

Generated answers and retrieval traces were frozen before judging. The automated judge evaluated each answer against the saved retrieved evidence and did not regenerate the answer. Judge outputs were treated as automated labels, not human expert validation. No inter-rater reliability statistic was calculated because there was no independent second human rater. Cohen's kappa was therefore not applicable.

Ethics and regulatory status

The study used public regulatory text files and generated benchmark questions. It did not involve human participants, patient data, animal studies, or a clinical trial. No ANVISA, regulatory authority, or journal endorsement is implied. The system is not intended to provide legal, clinical, or regulatory advice.

Results

Corpus and index characteristics

Cleaning reduced raw source files from large HTML-derived text exports to structured cleaned text files suitable for article-level indexing. The final index contained 382 detected articles and 481 chunks. Most articles were represented by one chunk; long articles were split while preserving article metadata. Chunk counts varied by source document, from 19 chunks for IN 338/2024 to 123 chunks for RDC 945/2024.

Internal held-out benchmark before final repairs

The first frozen held-out AI-judge evaluation included 150 rows, with 120 answerable and 30 unanswerable questions. Overall faithfulness on answerable rows was 0.997, citation correctness was 0.965, completeness was 0.807, and refusal correctness on unanswerable rows was 1.000. Language-stratified results showed a clear weakness in English completeness. English completeness was 0.645, while Portuguese completeness was 0.970. This result led to targeted review of English retrieval and answer completeness.

Post-repair internal held-out benchmark

After retrieval and completeness repairs, the English held-out run included 75 rows, with 60 answerable and 15 unanswerable questions. Operational source hit rate after reranking was 1.000, source MRR was 0.950, article Recall@5 was 1.000, article MRR@5 was 0.898, article

nDCG@5 was 0.924, and retrieved gold-term coverage was 0.994. Answer gold-term coverage was 0.849, citation presence was 1.000, article citation presence was 1.000, retrieved-citation validity was 1.000, and refusal accuracy was 1.000. AI-judge faithfulness was 0.997, citation correctness was 0.983, completeness was 0.995, and refusal correctness was 1.000.

The final Portuguese held-out run included 75 rows, with 60 answerable and 15 unanswerable questions. AI-judge faithfulness, citation correctness, completeness, and refusal correctness were all 1.000. These values should be interpreted cautiously. They were obtained after targeted repairs and therefore show that the final system passed the curated internal benchmark, not that general regulatory performance is perfect.

Table 2. Main post-repair internal held-out results.

Metric	English	Portuguese
Source hit after reranking	1.000	1.000
Article Recall@5	1.000	1.000
Answer term coverage	0.849	0.828
Retrieved-citation validity	1.000	1.000
Refusal accuracy	1.000	1.000
AI-judge faithfulness	0.997	1.000
AI-judge citation correctness	0.983	1.000
AI-judge completeness	0.995	1.000
AI-judge refusal correctness	1.000	1.000

Prospective external stress test

Because the internal metrics were high after repair, a 50-question prospective external stress test was run once before further tuning. The test contained 25 bilingual pairs, 40 answerable questions, and 10 unanswerable or adversarial questions. During artifact recovery, 40 preserved rows were combined with 10 regenerated unanswerable/adversarial rows using the same frozen settings. No retrieval, prompt, judge, threshold, or benchmark changes were made during recovery.

The external test supported answerable-question performance but exposed refusal weakness. Normalized article Recall@5 on answerable questions was 0.967 and normalized article MRR@5 was 0.912. Retrieved term coverage was 0.806, answer term coverage was 0.761, citation presence was 0.975, article citation presence was 0.975, and retrieved-citation validity was 0.975. AI-judge faithfulness was 1.000, citation correctness was 0.975, and completeness was 0.943. Refusal detection on unanswerable questions was 0.600 and AI-judge refusal correctness was 0.700.

By language, normalized article Recall@5 was 0.950 for English and 0.983 for Portuguese. AI-judge completeness was 0.930 for English and 0.955 for Portuguese. Refusal correctness was

0.800 for English and 0.600 for Portuguese. Error review showed that the system sometimes answered with nearby general rules when the corpus did not contain the exact requested fact.

Refusal-focused external test

After the first external test, refusal calibration was repaired using a separate development set. A new 30-question refusal-focused external test was then run. This test included 18 unanswerable or current-specific questions and 12 answerable controls.

On answerable controls, operational source hit after reranking was 1.000 and source MRR was 1.000. Retrieved gold-term coverage was 0.799. Citation presence, article citation presence, and retrieved-citation validity were all 1.000. AI-judge faithfulness and citation correctness were 1.000. Completeness was 0.908.

On unanswerable questions, refusal accuracy was 0.944, with a 95% confidence interval of 0.833 to 1.000. AI-judge refusal correctness was also 0.944. No over-refusal occurred on answerable controls. The remaining boundary case involved questions where the retrieved corpus provided a general enforcement consequence but did not provide the exact requested penalty.

Table 3. External testing results.

Metric	Prospective stress test	Refusal-focused external test
Questions	50	30
Answerable questions	40	12
Unanswerable/adversarial questions	10	18
Normalized article Recall@5, answerable	0.967	Not used because article labels were not normalized for this test
Retrieved-citation validity, answerable	0.975	1.000
AI-judge faithfulness, answerable	1.000	1.000
AI-judge citation correctness, answerable	0.975	1.000
AI-judge completeness, answerable	0.943	0.908
AI-judge refusal correctness, unanswerable	0.700	0.944
Over-refusal on answerable controls	Not reported	0.000

Discussion

Principal findings

This study developed and evaluated a bilingual article-aware RAG system for selected ANVISA regulatory documents. The final pipeline performed well on article-level retrieval, citation validity, answer faithfulness, completeness, and refusal behavior in internal testing. More importantly, external testing changed the interpretation of the project. The first prospective external test confirmed strong answerable-question behavior but showed that refusal calibration remained

the main weakness. After a separate refusal-development repair, the refusal-focused external test showed improved refusal correctness without over-refusal on answerable controls.

The main result is therefore not the presence of 1.000 internal scores. Those scores are useful only as evidence that the repaired system passed the curated internal benchmark. The more credible finding is the staged validation pattern: internal repair, untouched external stress testing, documented refusal failure, separate calibration, and a new external refusal test.

Comparison with prior regulatory RAG work

The closest published comparator is the TIRS study of semantic search of FDA guidance documents using generative AI [2]. That study evaluated a RAG application for FDA guidance questions and reported that correct source citation may help users find relevant guidance, but incorrect or incomplete answers remained a concern. The present study is aligned with that conclusion. It treats source and article citation as a primary endpoint and evaluates refusal behavior as a separate safety boundary.

This study differs in three main ways. First, the corpus was based on selected ANVISA regulatory documents rather than FDA guidance documents. Second, the evaluation was bilingual, with English questions over Portuguese source material. Third, the system was designed around article-level legal-regulatory structure rather than generic passage chunks. These differences make the work relevant to international regulatory affairs settings where users may need to search non-English health-authority documents while preserving source traceability.

Regulatory-science implications

The results support a limited use case: technical support for locating and summarizing evidence within a closed regulatory corpus. In that role, the system may help users identify relevant ANVISA documents and articles faster than manual search alone. The strongest design features for this use case were article-aware chunking, hybrid retrieval, reranking, bilingual query expansion, and citation-constrained generation.

The results do not support unsupervised regulatory decision-making. A RAG system can retrieve and summarize indexed text, but it does not determine current legal status, apply regulatory judgment to a full product dossier, or replace professional review. The closed-corpus design is important. Questions about current forms, fees, submission portals, private company status, future regulatory changes, or documents outside the index should be refused.

Refusal calibration

Refusal behavior was the main safety boundary. The first external test showed that the system sometimes treated related context as sufficient even when the exact requested fact was absent. This is a common regulatory risk. A nearby article may discuss the same topic but not answer the specific question. In such cases, a fluent answer can be more harmful than a refusal because it may appear authoritative.

The refusal-focused repair improved refusal correctness from 0.700 in the first external stress test to 0.944 in the second refusal-focused external test. The repair also preserved answerability, with no over-refusal on answerable controls. This does not prove that refusal is solved. It shows that refusal should be evaluated separately from answer correctness and should remain a monitored endpoint in future work.

Automated evaluation

The project used an AI judge because a two-expert panel was not available. This choice makes the study feasible as a solo technical validation project, but it limits the claim level. Automated judging can consistently apply a rubric for faithfulness, citation correctness, completeness, and refusal correctness. It can also score larger benchmarks than a single manual reviewer can practically assess. However, it may miss domain-specific legal nuance, may favor answers that match its preferred style, and cannot establish expert consensus.

For that reason, the manuscript should describe AI-judge scores as automated technical labels. It should not report inter-rater reliability, expert agreement, or regulatory correctness. Future validation should compare automated scores against independent regulatory reviewers and include disagreement analysis.

Limitations

The study has several limitations. First, the corpus was closed and selected. It does not represent all ANVISA regulations, current web pages, forms, fees, submission systems, or future regulatory updates. Second, the benchmark was created without an independent regulatory expert panel. Third, answer-quality scoring used one automated judge rather than multiple human reviewers. Fourth, internal results may be inflated because the pipeline was repaired after error analysis. Fifth, the first external stress test required artifact recovery after output files were moved; the recovery preserved settings, but the caveat should be reported. Sixth, the system may still fail on boundary questions where the retrieved text is related but

incomplete. Seventh, the system was tested as a technical prototype, not as a deployed regulatory affairs tool.

Future work

Future work should expand the corpus, add version tracking for regulatory updates, test additional languages, and compare AI-judge scores with independent regulatory reviewers. External benchmarks should be authored by people not involved in system development. Evaluation should include time-to-answer studies, user trust calibration, and controlled comparisons against keyword search. Refusal testing should remain separate from ordinary answerable-question testing.

Conclusion

A bilingual article-aware RAG pipeline can retrieve and summarize selected ANVISA regulatory documents with source and article citation when the corpus contains the answer. Hybrid retrieval, reranking, Portuguese-English query expansion, citation constraints, and refusal calibration were all needed to reach stable technical performance. The system should be presented as a regulatory information-access prototype with transparent limits, not as regulatory advice or expert validation.

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Data Availability

The cleaned corpus files, benchmark manifests, evaluation outputs, prompts, and code used for the technical validation are available in the project repository or can be provided as

supplementary material, subject to any redistribution limits for source regulatory text. Benchmark and output files were saved with manifests and checksums to support reproducibility.

Ethics Approval

Ethics approval was not required because the study used public regulatory documents and generated benchmark questions. No human participants, patient data, animal data, or clinical trial data were used.

Clinical Trial Registration

Not applicable.

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